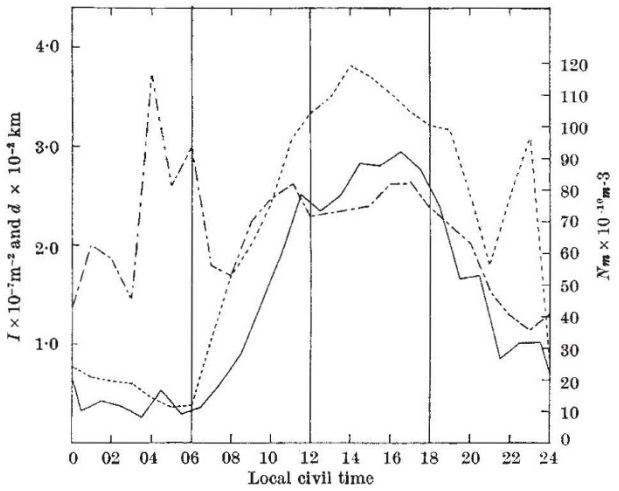


Fig. 2. Diurnal variation of total electron content (I), maximum electron density (Nm) and equivalent slab thickness (d) of the ionosphere over Nairobi, for the period Oct. 1964 to Jan. 1965. ——, I; - - - -, Nm; —·—·—, d

which is located on the magnetic equator, a similar, though smaller, anomalous increase in equivalent slab thickness around layer sunrise is indicated and attributed to a departure from thermal equilibrium. In the same report, the mean diurnal range of total electron content is given as 10 : 1, and shows a rise starting at about 10·00 local time to a maximum at about 18·00. The values of I obtained there are of the same order of magnitude as presented here.

It is hoped shortly to publish more detailed results on total electron content at this station. This work was supported by the Rockefeller Foundation and by the loan of two receivers from the U.S. Air Force Cambridge Research Laboratories.

A. N. Hunter
A. R. Webster
R. F. Kelleher
P. E. Miall

University College Nairobi,
(University of East Africa),
Nairobi, Kenya.

[1] Blackband, W. T., *J. Geophys. Res.*, **65**, 1987 (1960).
[2] Garriott, O. K., *J. Geophys. Res.*, **65**, 1151 (1960).
[3] Browne, I. C., *et al.*, *Proc. Phys. Soc.*, B, **69**, 901 (1956).
[4] Blumle, L. J., *J. Geophys. Res.*, **67**, 4601 (1962).

Palaeomagnetic Measurements on some Karroo Dolerites from Rhodesia

In a previous investigation of the Pre-Cambrian dolerites of Eastern Rhodesia, McElhinny and Opdyke[1] established on the basis of palaeomagnetic measurements that two distinct periods of igneous activity were represented. At one locality it appeared that the Mashonaland dolerites had intruded the Umkondo dolerites. On the basis of the evidence from this one site they suggested that the Umkondo dolerites were the older of the two sets of intrusions.

The results of some other palaeomagnetic measurements being made in this Department, together with a number of discordant potassium–argon dates from specimens of the Umkondo dolerites, necessitated placing the evidence for the sequence of the two sets of intrusions on a firmer basis. In one part of the Middle Sabi Valley, Swift[2] reports that the main Umkondo dolerite sill has been intruded by a large dyke of much finer-grained dolerite, petrologically very similar to the main sill. The possibility existed that this represented a second instance where the Mashonaland dolerite had intruded the Umkondo dolerite. Further south in the Umkondo Copper Mine, we discovered from the management that three ages of dolerite intrusion had been identified in the mine workings. A dolerite sill has intruded the Umkondo sediments and this sill is cut by a dyke. Both the sill and the dyke are then cut by a second dyke. This situation represented yet a further possibility of establishing beyond doubt the relationship between the Mashonaland and Umkondo dolerites.

Three blocks, oriented with a compass, were collected from each of the two dykes and the sill underground in the Umkondo mine workings, taking care to keep away from the baked contacts. Four blocks, similarly oriented, were collected from a stream section in the fine-grained dyke reported by Swift[2]. From each block two cylinders 1 in. in diameter and 1 in. long were drilled and the natural remanent magnetization was measured on the spinner magnetometer described by Gough[3]. At each site the natural remanent magnetization formed good groups. From three of the sites at least one core from each of the sites was demagnetized successively in alternating magnetic fields of up to 60 milliweber m^{-2} peak value, using the apparatus described by McElhinny[4]. No significant change in the direction of magnetization was observed in any case. This indicates that the natural remanent magnetization at these sites is stable and probably represents the thermoremanence acquired at the time of their intrusion. The specimens from the last intruded dyke of the Umkondo mine were so weakly magnetized that it was not possible to conduct demagnetization experiments on them. The parameters of the natural remanent magnetization at the four sites are given in Table 1.

The directions of magnetization of the four sites, which are very similar, correspond neither with that of the Mashonaland dolerite nor with that of the Umkondo dolerite. The investigation thus failed to establish the sequence of the intrusions of the Mashonaland and Umkondo dolerites.

The directions observed correspond closely to those observed in the Karroo dolerites[5,6]. Since Karroo sediments occur in the region between the two areas sampled, it is reasonable to suppose that we, in fact, sampled Karroo dolerites, which are known to occur only 25 km further south.

Graham and Hales[5] were the first to carry out palaeomagnetic measurements on Karroo dolerites. Whereas their collection of surface samples in South Africa show considerable scatter in direction, samples from four sites in underground exposures exhibited consistent directions of magnetization. There is some evidence for the stability of these samples from baked contact investigations even though no alternating field demagnetization tests were carried out. In addition, two of the sites exhibit normal polarity, while the other two exhibit reversed polarity with almost 180° separating the two directions. This provides further evidence for their stability. In addition Nairn[6] has published results from a Karroo dolerite near Durban and from a limburgite plug in Rhodesia. In the former case, alternating field demagnetization was undertaken on the samples.

Since the palaeomagnetic pole position recalculated by Cox and Doell[7] from Graham and Hales's four sites has a large angle of 95 per cent confidence (20°), it seems worth while collecting together the results at present available from ten sites, including the four presented here, to calculate a new pole for the Karroo dolerites. The parameters of the mean of the ten site pole positions are shown in Table 2, and it is thought that this now represents a better estimate of the palaeomagnetic pole for the Karroo dolerites than was previously available.

Table 1

Site	Lat. S.	Long. E.	N	D	I	k
Middle Sabi Valley	19° 38′	32° 22′	4	319·3°	−59·3°	170
Umkondo Mine, sill	20° 19′	32° 09′	3	318·5°	−35·7°	18·4
Umkondo Mine, 1st dyke	20° 19′	32° 09′	3	319·8°	−58·6°	34·8
Umkondo Mine, 2nd dyke	20° 19′	32° 09′	3	8·1°	−57·7°	7·30

N, number of samples; D, declination; I, inclination; k, precision parameter.

©1965 Nature Publishing Group

Table 2

N	R	K	Lat. S.	Long. E.	A_{95}
10	9·451	16·4	65·4°	75·1°	12·3°

N, number of sites; R, resultant of N unit vectors; K, precision of site poles; A_{95}, semi-angle of the cone of 95 per cent confidence.

McDougall[8] has made potassium–argon age determinations on a number of Karroo dolerites from South Africa and finds that they have a spread of ages from 154 to 190 million years.

We thank the manager and staff of the Umkondo Mine for their help in collecting the samples from the mine.

M. W. McElhinny
D. L. Jones

Department of Physics,
University College of Rhodesia and Nyasaland,
Salisbury.

[1] McElhinny, M. W., and Opdyke, N. D., *J. Geophys. Res.*, **69**, 2465 (1964).
[2] Swift, W. H., *Southern Rhodesian Geol. Surv. Bull.* No. 52, 17 (1962).
[3] Gough, D. I., *J. Geophys. Res.*, **69**, 2455 (1964).
[4] McElhinny, M. W., *Geophys. J.* (in the press).
[5] Graham, K. W. T., and Hales, A. L., *Phil. Mag.*, Suppl., **6**, 149 (1957).
[6] Nairn, A. E. M., *Overseas Geol. and Mineral Resources*, **9**, 302 (1964).
[7] Cox, A., and Doell, R. R., *Bull. Geol. Soc. Amer.*, **71**, 645 (1960).
[8] McDougall, I., *J. Geophys. Res.*, **68**, 1535 (1963).

PHYSICS

Low-energy Electron–electron Bremsstrahlung

The emission of radiation in the scattering of electrons by atoms, free electrons, or nuclei generally is considered to be well understood. In fact, extensive reviews of bremsstrahlung cross-sections are available for both the non-relativistic[1,2] and relativistic domains[2,3]. Nevertheless, there are some unresolved problems concerning the radiation emitted in electron–electron collisions—that is, collisions in which momentum is absorbed by a free electron or through the excitation or ejection of a bound electron.

At relativistic energies the cross-section for electron–electron bremsstrahlung is found to be comparable with that for electron–proton bremsstrahlung[4,5]. With decreasing energy in the non-relativistic domain, however, the cross-section is supposed to decrease rapidly relative to the electron–proton bremsstrahlung cross-section due to the zero dipole moment of the two-electron system. Thus radiative electron–electron collisions are usually ignored in determinations of such quantities as the opacity of a gas due to free–free transitions, the intensity of the X-ray continuum, or the emission of radiation from an ionized gas. In this communication, I shall point out some discrepancies in this low-energy domain. My interest stems not only from the possible relevance of electron–electron bremsstrahlung to these practical problems but also from its possible relevance to the radiative reaction problem. Effects due to atomic binding are not at issue, so that I consider the target electron to be free throughout this discussion.

The conclusion that electron–electron bremsstrahlung is negligible in the low-energy limit stems primarily from the classical result for the energy radiated in the collision of two charged particles. For particles with charge e and masses m and M, this cross-section is given in the dipole approximation by[6]:

$$\mathrm{d}\sigma\,(cl,\ \text{dipole}) = \frac{16}{3}\alpha r_0^2\left(\frac{c}{v}\right)^2\left(1-\frac{m}{M}\right)^2\left[\ln\frac{2\mu v^3}{e^2\omega}-0{\cdot}57\right]\frac{\mathrm{d}\omega}{\omega} \qquad (1)$$

For $M \to \infty$ this result differs little from the non-relativistic limit of the Bethe–Heitler formula. For $M = m$ this classical dipole contribution vanishes, so that the cross-section for radiation is given approximately by the quadrupole contribution. Apart from a numerical factor, this is:

$$\mathrm{d}\sigma\,(cl,\ \text{quadrupole}) \approx \frac{e^6}{m^2v^2c^3}\frac{\mathrm{d}\omega}{\hbar\omega}\times\left(\frac{\omega a}{c}\right)^2 \approx \alpha r_0^2\left(\frac{\omega}{\omega_0}\right)^2\frac{\mathrm{d}\omega}{\omega} \qquad (2)$$

where $a \equiv e^2/mv^2$ and $\omega_0 \equiv (e^2/ma^3)^{1/2}$ are the characteristic length and frequency for a Coulomb collision[7]. We see that the ratio of this cross-section to the cross-section for dipole radiation in electron–proton scattering is of the order of:

$$R \approx \left(\frac{\omega}{\omega_0}\right)^2\left(\frac{v}{c}\right)^2 \approx \left(\frac{e^2}{\hbar c}\right)^2\left(\frac{\omega}{\omega_{\max}}\right)^2 \qquad (3)$$

Here $\omega_{\max}$ is defined by the (quantum) condition $\hbar\omega_{\max} \approx mv^2$.

The status of the quantum theory of non-relativistic bremsstrahlung is much less clear. Katzenstein[8] found that the low-frequency limit vanished in agreement with the classical theory. Redhead[9], however, found a number of corrections to this calculation which, when taken into account, yielded:

$$\mathrm{d}\sigma\,(qu.) \approx \frac{32}{5}\alpha r_0^2\frac{\mathrm{d}\omega}{\omega}\left(\log\frac{4mv^2}{\hbar\omega}-\frac{4}{5}\right) \qquad (4)$$

Although this result is smaller than the electron–proton cross-section by a factor $(v/c)^2$, it has the same frequency dependence and thus diverges in the low-frequency limit. This behaviour differs markedly from the classical results given in equations (1) and (2) where the bremsstrahlung cross-section for identical particles clearly vanishes in the low-frequency limit.

This reaction has also been investigated by Fedyushin[10] and Garibyan[11] and discussed by Joseph and Rohrlich[2]. They find the same numerical coefficient and frequency dependence as Redhead but an extra factor of $(c/v)^2$:

$$\mathrm{d}\sigma\,(qu.) \approx \frac{32}{5}\alpha r_0^2\left(\frac{c}{v}\right)^2\frac{\mathrm{d}\omega}{\omega}\left(\log\frac{4mv^2}{\hbar\omega}+\frac{17}{12}\right) \qquad (5)$$

Fedyushin's derivation is semi-classical and entails an explicit evaluation of the quadrupole moment. Garibyan, on the other hand, obtained this result by a quantum theoretic calculation, the details of which apparently are not available[12]. (The matrix elements for electron–electron bremsstrahlung have also been derived according to quantum electrodynamics by Hodes[13] and Gluckstern[14], but these results have not been evaluated in the non-relativistic limit.) Due to the factor $(c/v)^2$, the discrepancy between equation (5) and the classical cross-sections for electron–electron bremsstrahlung is even greater than that between equation (4) and the classical results. In fact, not only does this cross-section depend on frequency and electron energy exactly as does the non-relativistic cross-section for electron–proton bremsstrahlung but it has a slightly larger numerical coefficient as well.

If we assume that this last result (equation (5)) is correct, it is clear that electron–electron bremsstrahlung is a more significant process than generally believed. In ionized hydrogen the opacity arising from free–free transitions will be almost doubled by the additional absorption by electron pairs. Production of continuous X-rays will be enhanced at non-relativistic as well as relativistic energies by the approximate factor $(Z+1)/Z$. The emission of thermal bremsstrahlung from optically thin plasmas is similarly increased.

There are, however, a number of reasons to believe that equation (5) is wrong despite its wide[2,3] acceptance. Chief

©1965 Nature Publishing Group